Plastic Cup Manufacturing Material Waste Analysis

Plastic cup manufacturing may waste 13% of materials due to trimming losses and production inefficiencies, this should be less than 10%

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Executive Summary

ACTUAL LOSS RATE VS. TARGET

13% vs. **10% Max Target**

Resulted in **17,959.04 kg** of excess material loss above allowable limits over two years

Total Resin Consumed

600,000 kg

Total Material Scrap

78,000 kg

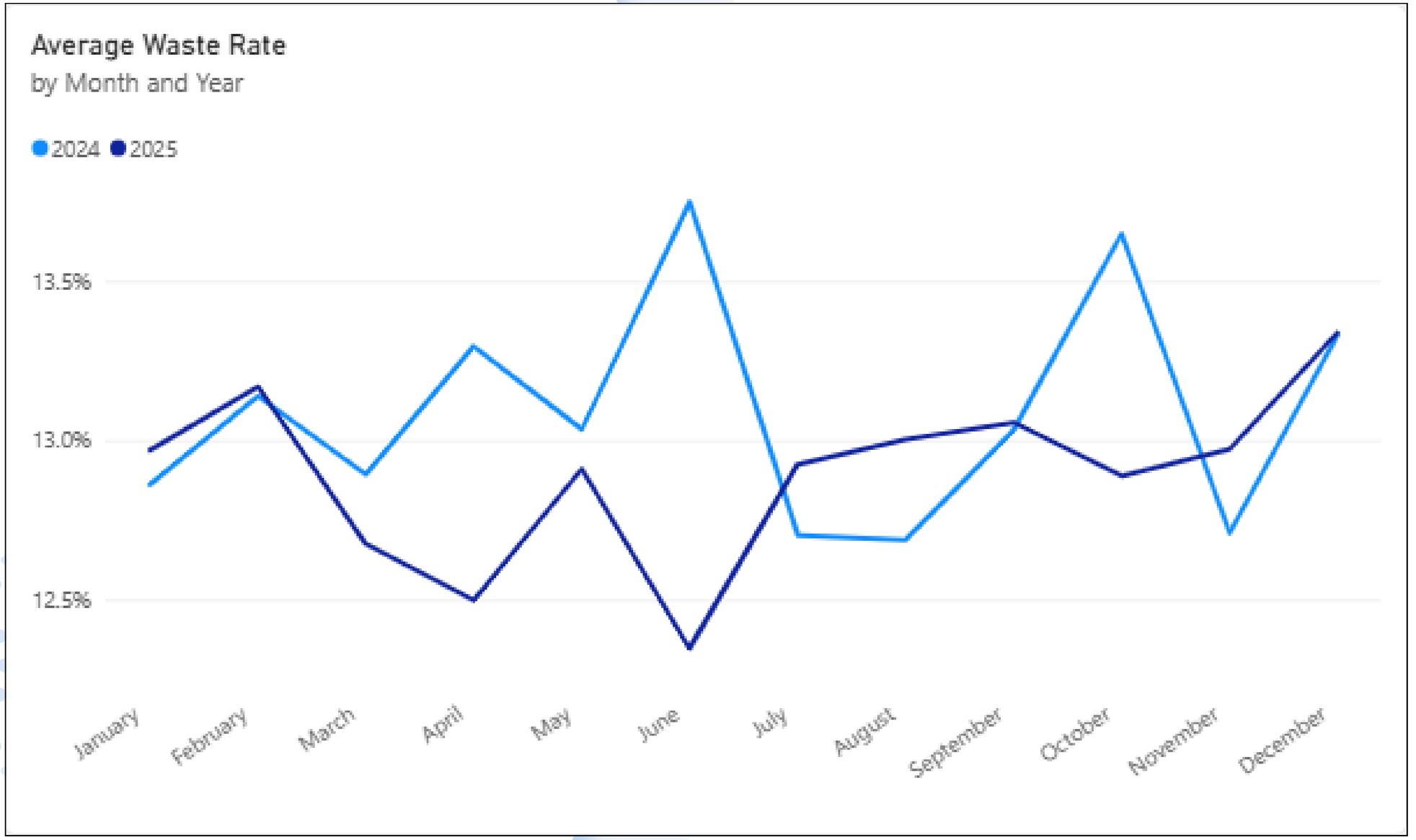
- Operational performance across 731 continous days confirms levels are structurally locked around 13%
- Achieving the <10% threshold will directly recover almost **18 tons** of prime resin material

Annual Production & Waste Metrics (2024 vs. 2025)

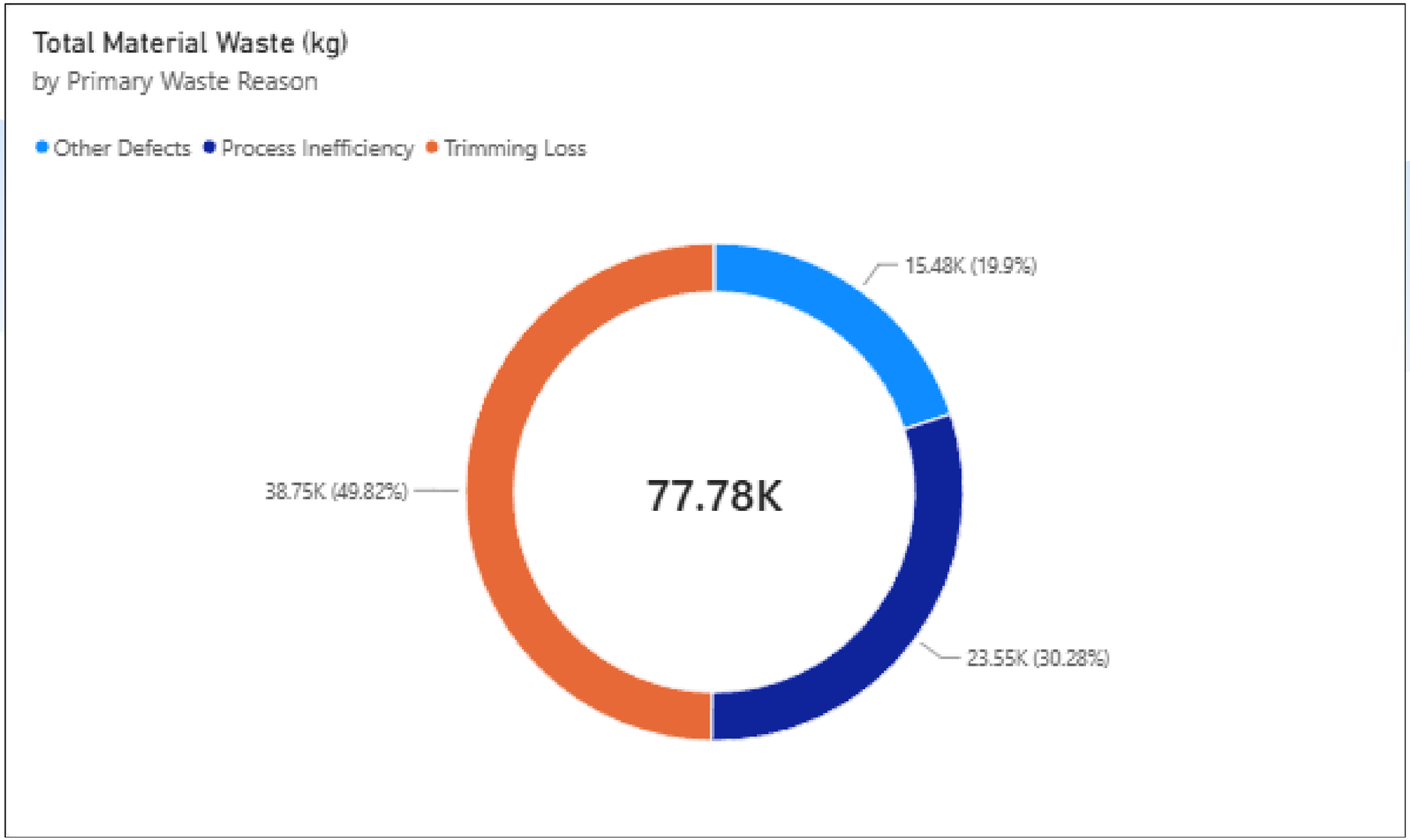
2024	Total Material Used (kg) 301.48K	Total Material Waste (kg) 39.34K	Overall Waste Rate (%) 13.09%
2025	Total Material Used (kg) 299.16K	Total Material Waste (kg) 38.44K	Overall Waste Rate (%) 12.90%

Data Cleaning

Monthly Average Waste Rate Trend Analysis (2024 vs. 2025)



Total Material Waste Breakdown



Key Findings: Scrap Driver Hierarchy

PRIMARY DRIVER

49.82%

Trimming Loss

38,751.70 kg total scrap

SECONDARY DRIVER

38.28%

Process Inefficiency

23,554.20 kg total scrap

QUALITY LOSS

19.90%

Other Defects

15,478.30 kg total scrap

Strategic Recommendations to Reach <10.0% Waste Target

PILLAR 1 - TRIMMING

Tooling & Cavity Layouts

Re-design web spacing and mold geometry

Target: 4.5% - 5% Trimming scrap

PILLAR 2 - PROCESS

TPM on Heating & Molds

Prevent drift in production quality and machine downtime scrap

Target: 2.5% - 3% Process Loss

PILLAR 3 - QUALITY

Real-Time Defect Detection

Integrate optical inline inspection systems

Target: 1.5% - 2% Quality Defects

Thank You

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